

Множества и отношения

Задание 1.

Задача 1.1.

$$A = \{1, 2, 3, 4, 5, 6, 7\} \quad C = \{2, 4, 6, 8, 10\}$$

$$B = \{4, 5, 6, 7, 8, 9, 10\} \quad U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

$$a) A \cup C = \{1, 2, 3, 4, 5, 6, 7, 8, 10\}$$

$$б) A \cap B = \{4, 5, 6, 7\}$$

$$в) A \cap (B \cup C) = \{2, 4, 5, 6, 7\}$$

$$1) B \cup C = \{2, 4, 5, 6, 7, 8, 9, 10\}$$

$$2) (A \cap B) \cup C = \{2, 4, 5, 6, 7, 8, 10\}$$

$$1) A \cap B = \{4, 5, 6, 7\}$$

$$г) (A \cap B)' = U \setminus (A \cap B) = \{1, 2, 3, 8, 9, 10\}$$

$$е) A' \cap B' \neq$$

$$1) A' = U \setminus A = \{8, 9, 10\}$$

$$2) B' = U \setminus B = \{1, 2, 3\}$$

$$\Rightarrow A' \cap B' = \emptyset$$

$$ж) A \Delta B = \{1, 2, 3, 8, 9, 10\}$$

$$з) A - B = \{1, 2, 3\}$$

Задача 1.2

$$A = \{1, 2, 3, 4, 5, 6, 7\} \quad C = \{2, 4, 6, 8, 10\}$$

$$B = \{4, 5, 6, 7, 8, 9, 10\} \quad U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

a) $A - C = \{1, 3, 5, 7\}$

б) $(A - B) \cup (B - A) = \{1, 2, 3, 8, 9, 10\}$

1) $A - B = \{1, 2, 3\}$

2) $B - A = \{8, 9, 10\}$

в) $A \cap (B \cap C') = \{5, 7\}$

1) $C' = U \setminus C = \{1, 3, 5, 7, 9, 10\}$

2) $B \cap C' = \{5, 7, 9\}$

г) $(A \cup C) - B' = \{4, 5, 6, 7, 8, 10\}$

1) $A \cup C = \{1, 2, 3, 4, 5, 6, 7, 8, 10\}$

2) $B' = U \setminus B = \{1, 2, 3\}$

д) $(A - \emptyset) \cup (A - A) = \{1, 2, 3, 4, 5, 6, 7\}$

1) $A - \emptyset = A$

2) $A - A = \emptyset$

е) $B \Delta C = \{2, 5, 7, 9\}$

ж) $C - A = \{8, 10\}$